CELEMICS

Kit User Manual

NGS Library Preparation

Enzymatic Preparation Kit (EP-Kit)

—

Illumina Platform
Unique Dual Index

Research Use Only

CELEMICS

Materials

Library Preparation #1 **Store at -20°C**

Box No.	Product Name	1 rxn (µL)	Cat No. 16 samples	Cat No. 32 samples	Cat No. 48 samples	Cat No. 96 samples
LI11	EP-ER/A Enzyme	6	LI11S-N-F00-16	LI11S-N-F00-32	LI11S-N-F00-48	LI11S-N-F00-96
	EP-ER/A Buffer	3				
	Ligation Enzyme	4				
	Ligase Buffer	12				

Library Preparation #2 **Store at -20°C**

Box No.	Product Name	1 rxn (µL)	Cat No. 16 samples	Cat No. 32 samples	Cat No. 48 samples	Cat No. 96 samples
LI22	UDI-Primer	5	LI22S-S-800-16	LI22S-S-800-32	LI22S-S-800-48	LI22S-S-800-96
	Short Adapter	10				

Additional Accessory Materials

Additionally Provided Materials List

Product Name	Cat No.	Store at (°C)
CeleMag Clean-up Bead	CMCB-M-050 (5 mL) CMCB-M-500 (50 mL)	4°C
CLM Polymerase	CMPF-M-500 (5 mL)	-20°C

* The All-in-One Package includes additional accessory materials with dedicated capacity according to the package option, so it may differ from the Cat No. provided in the table above.

Materials Required (Not Provided)

Not Provided Materials List

Reagents & Consumables		
Product Name	Suggested Supplier	p/n
Nuclease-free water (not DEPC-treated)	General lab supplier	-
Ethanol (200 proof)	General lab supplier	-
Conical centrifuge tubes (15 mL or 50 mL)	General lab supplier	-
P10, P20, P200, and P1000 pipettes	General lab supplier	-
Multichannel pipette	General lab supplier	-
Sterile, nuclease-free aerosol barrier pipette tips	General lab supplier	-
0.2 mL PCR Tube Strips, Low Profile and Cap Strips	General lab supplier	-
Powder-free gloves	General lab supplier	-
DNA LoBind Tubes, 1.5 mL PCR clean	Eppendorf	022431021 or equivalent
Microtubes 1.5ml Clear, Sterile	AXYGEN	MCT-150-C-S or equivalent
Eppendorf™ twin.tec™ 96-well LoBind PCR Plates, Skirted	Eppendorf	0030129512 or equivalent
Microseal 'B' adhesive seals	Bio-Rad	MSB-1001 or equivalent

Equipment		
Product Name	Suggested Supplier	p/n
TapeStation Instrument	Agilent Technologies, Inc.	G2991BA or equivalent
Magnetic separator	Invitrogen	12321D or equivalent
Thermal Cycler	General lab supplier	-
Micro-centrifuge	General lab supplier	-
Vortex mixer	General lab supplier	-

Before You Begin

Before You Begin

It is recommended to read the manual carefully to understand the experimental procedure. Please follow the text **in bold** and the text highlighted in the right "NOTE" box carefully. The number in the "NOTE" box matches the procedure on the left.

Notice

Experimental results may be affected by DNA quality. We recommend using high-quality DNA samples. Low-quality samples (e.g., FFPE tissue extracted gDNA) show variable efficiency due to chemical damage and fragmentation, and its small amounts. Therefore, depending on the sample quality and input amount, the protocol may require additional experimental optimization.

Directions for the Use of This Product

1. This product should be used for research use only.
2. Reagents and specimens should be handled cautiously to avoid potential injury and infections.
    1) Handle reagents cautiously and avoid contact with any part of the body, especially skin, eyes, or mucous membranes. In the event of reagent contact, rinse immediately with water and follow duly accredited treatment.
    2) All blood serum materials from human sources should be handled carefully and treated as potential infectious sources.
3. Genomic DNA extracted from bio-specimens such as blood, saliva, FFPE, and fresh tissue should be used as sample specimens.
4. To prevent cross-contamination, change the pipette tip between each sample or reagent.
5. Usage of reagents should be followed by the instruction. We would recommend not using reagents past their expiration dates.
6. Reagents should be stored at their suggested conditions.
    Room temperature (RT, 15-25°C), Freezer(-20°C), Deep freezer (-80°C)
7. The viscosity of the reagent is different. Slow pipetting may help with accuracy when handling viscous reagents.
8. For disposal of used or expired reagents, please contact an authorized waste disposal company in accordance with relevant laws and regulations.
9. Be cautious with specific allergenic reagents.

Please maintain a nuclease-free environment during the experiment. Wear a lab coat, disposable gloves, and protective goggles. If you have any questions or concerns, please email us at manual@celemics.com

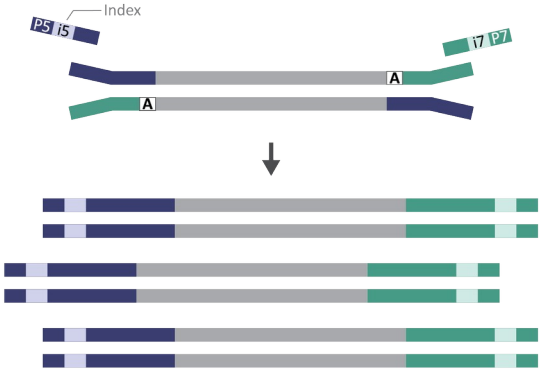
Protocol Overview

Genomic DNA

Enzymatic preparation
1 hr

Adapter ligation
35 mins

Amplification of the adapter-ligated library
20 min



Purification of sample using CeleMag Clean-up Beads
Assess quality and quantity
30-60 min

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NGS Library Preparation

Step 1. Enzymatic Preparation

This step is to construct libraries with an average insert size of ~300 bp using genomic DNA inputs ≥ 50 ng for targeted enrichment and direct sequencing, respectively.

EDTA-containing samples are not suitable for this step. See the appendix before starting and have a buffer exchange process.

The fragmentation time provided is compatible with high-quality samples. Optimization for fragmentation time may be required for low-quality samples (e.g., FFPE). See the appendix for more information about FFPE optimization.

[Preparation]

- **Library Preparation Box #1** – Place on ice (or use a cooling block)
  EP-ER/A Buffer
  EP-ER/A Enzyme

[Procedure]

1. Adjust the DNA sample into a new PCR tube (or 96-well plate) as shown. The amount of DNA should be ≥ 50 ng.

Adjust Volume per Sample

Reagent	Volume for 1 Sample (µL)
DNA sample	x
Nuclease-free water	21 - x
Total Volume	**21**

2. Prepare Enzymatic Prep Master Mix described below.

Enzymatic Prep Master Mix

Reagent	Volume for 1 Sample (µL)	Volume for 8 Samples (µL) (with 10% overage)
EP-ER/A Buffer	3	26.4
EP-ER/A Enzyme	6	52.8
Total Volume	**9**	79.2

3. **Add 9 µL of Enzymatic Prep Master Mix** to each DNA sample tube. Mix well using a vortex mixer, and pulse spin down the tube.

NOTE

1. If you prefer to proceed with less than 50 ng, please contact us at manual@celemics.com

2. Mix Enzymatic Prep Master Mix thoroughly.

3. Make sure all sample-mixture tubes are placed on ice. Enzymes may work at room temperature and fragment DNA to undesired sizes.

4. Place the tube (30 µL) on a thermal cycler and run the following program.

Thermal Cycler Program (lid temperature: 70°C)

Step	Temperature	Time
1	4°C	1 minute
2	32°C	Variable *
3	65°C	30 minutes
4	4°C	Hold

⚠ **IMPORTANT:** Proceed immediately to the next step. **

NOTE

* The fragmentation times differ between enzyme production lots. Please find the library size table based on fragmentation time on the EP kit webpage. (The table is based on the fragmentation of 100 ng DNA with DIN≥9.) https://celemics.com/en/sub/search/CoA_of_Celemics_EP_Kit.asp

** ER/A-tailed samples can be kept at 4 °C for **no more than one hour.**

4-1. Desired time for fragmentation depends on the amount of input DNA. See 'Appendix: Optimize Celemics kit protocol based on your samples.'

4-2. This kit can be applied to damaged samples such as FFPE. However, it should be optimized to find an appropriate fragmentation time. See 'Appendix Optimize Celemics kit protocol based on your samples' for more information on fragmentation time.

Step 2. Adapter Ligation

[Preparation]

- **Library preparation Box #1 -** Place on ice (or use a cooling block)
  Ligation Buffer
  Ligation Enzyme
- **Library preparation Box #2 -** Thaw on ice
  Short Adapter

IMPORTANT:
If less than 50 ng of initial DNA is used, 1/10 dilution of the Short Adapter may help improve library preparation efficiency. Please refer to the technical note for more details.

[Procedure]

1. **Add 10 µL of Short Adapter** directly into the ER/A-tailed sample.
2. Mix well using a vortex mixer, and pulse spin down the tube.
3. Prepare Ligation Master Mix described below.

Ligation Master Mix

Reagent	Volume for 1 Sample (µL)	Volume for 8 Samples (µL) (with 10% overage)
Nuclease-free water	4	35.2
Ligation Buffer	12	105.6
Ligation Enzyme	4	35.2
Total Volume	**20**	176

4. **Add 20 µL of Ligation Master Mix** into the ER/A-tailed & Adapter sample.
5. Mix well using a vortex mixer, and pulse spin down the tube.
6. **Incubate the tube (60 µL) at 20°C for 20 minutes (lid temperature: 40°C).**

IMPORTANT: Proceed immediately to the next step.

NOTE

3. The viscosity of reagents is high, mix well using vortex mixer.

5. Make sure all sample-mixture tubes are placed on ice.

Step 3. Purification of Sample Using CeleMag Clean-up Bead

[Preparation]

- **CeleMag Clean-up Bead -** Place the bead at room temperature for at least 30 minutes before use. Store the beads at 4°C after use.
- **80% ethanol solution -** Using freshly made 80% ethanol solution is recommended.

Volume of CeleMag Clean-up Bead

Reagent	Volume Ratio (Bead : Sample)
CeleMag Clean-up Bead	**1 : 1** (e.g., 60 µL : 60 µL)

[Procedure]

1. Vortex CeleMag Clean-up Bead to achieve a homogeneous state.
2. **Add 60 µL CeleMag Clean-up Bead** to a new 1.5 mL LoBind Tube (0.2 mL PCR tube or 96-well plate is also available) and **transfer the Adapter ligated DNA (60 µL**) to the tube.
3. Mix well using a vortex mixer and pulse spin down the tube.
4. Incubate at room temperature for **5 minutes**.
5. Place the tube on the magnetic separator until the solution is clear (3-5 minutes).
6. Keep the tube on the magnetic separator and **discard the supernatant**.
7. While keeping the tube on the magnetic separator, **add 500 µL** of **fresh 80% ethanol** to each tube (180 µL for PCR tube or 96-well plate).
8. Incubate for 30 seconds at room temperature, then **discard ethanol**.
9. **Repeat wash** again (step 7-8).
10. Spin down, place the tube on the magnetic separator, and **discard residual ethanol**.
11. **Air-dry the sample tube** at room temperature for **2 minutes** or until residual ethanol is completely evaporated.
12. **Add 22 µL of nuclease-free water** to the sample tube and vortex to mix. Pulse spin down and **incubate for 2 minutes** at room temperature.
13. Place the tube on the magnetic separator for **2 minutes** until the solution is clear.
14. Transfer the supernatant (20 µL) to a new tube.

STOP POINT:
If you are not immediately continuing to the next step, store the sample at -20°C.

NOTE

3. To avoid the bead precipitation, we do not recommend spinning down the CeleMag Clean-up bead too strongly.

6. Do not disturb the bead pellet while you remove the supernatant.

11. Overdrying of the bead pellet will decrease the elution efficiency.

Step 4. Amplification of the Adapter-Ligated Library

[Preparation]

- **CLM Polymerase -** Place on ice (or use a cooling block)
- **Library Preparation Box #2** – Thaw on ice, and keep the reagents on ice
  UDI-Primer

[Procedure]

1. **Add 25 µL of the CLM Polymerase** to the new PCR tube.
2. **Add 5 µL of UDI-Primer** to the individual PCR tube.
3. **Add Adapter-ligated DNA library (20 µL)** to the PCR tube.
4. Mix well using a vortex mixer, and pulse spin down the tube.
5. Place the tube (50 µL) on a thermal cycler and run the PCR program.

PCR Program

Temperature	Time	Number of Cycles
98°C	45 seconds	1
98°C	15 seconds	See **'The Number of PCR Cycles'**
65°C	15 seconds	
72°C	30 seconds	
72°C	3 minutes	1
4°C	Hold	-

The Number of PCR Cycles

gDNA Input Amount (ng)	Total PCR Cycles
500 ng ≤ gDNA amount	3-4
100 ng ≤ gDNA amount < 500 ng	5-6
50 ng ≤ gDNA amount < 100 ng	7-8
gDNA amount < 50 ng	9-10

⚠ **IMPORTANT:** Proceed immediately to the next step right away.

⚠ **IMPORTANT:** If you want to proceed with normalization using NM Beads, please follow the **'CeleNM Bead - Purification & Quantification Protocol'** after completing the 'Amplification of the Adapter-Ligated Library' step.

NOTE

1. Make sure all sample-mixture tubes are placed on ice.

2-1. If using a 96-well plate format of UDI-Primer, spin down before use.

2-2. Record the appropriate UDI-Primer per sample.

2-3. Use different UDI-Primer when multiple samples run in the same sequencing run.

5-1. The quality of gDNA can affect PCR efficiency.

5-2. The number of PCR cycles depends on the amount and quality of input DNA. See 'Appendix: Optimize Celemics kit protocol based on your samples' to optimize.

Step 5. Purification of Sample Using CeleMag Clean-up Bead

[Preparation]

- **CeleMag Clean-up Bead -** Place the beads at room temperature for at least 30 minutes before use. Store the bead at 4°C after use.
- **80% ethanol solution -** Using freshly made 80% ethanol solution is recommended.

Volume of CeleMag Clean-up Bead

Reagent	Volume Ratio (Bead : Sample)
CeleMag Clean-up Bead	**1 : 1** (e.g., 50 µL : 50 µL)

[Procedure]

1. Vortex CeleMag Clean-up Bead to achieve a homogeneous state.
2. **Add 50 µL CeleMag Clean-up Bead** to a new 1.5 mL LoBind Tube (0.2 mL PCR tube or 96-well plate is also available) and **transfer the amplified sample (50 µL**) to the tube.
3. Mix well using a vortex mixer and pulse spin down the tube.
4. Incubate at room temperature for **5 minutes**.
5. Place the tube on the magnetic separator until the solution is clear (3-5 minutes).
6. Keep the tube on the magnetic separator and **discard the supernatant**.
7. While keeping the tube on the magnetic separator, **add 500 µL** of **fresh 80% ethanol** to each tube (180 µL for PCR tube or 96-well plate).
8. Incubate for 30 seconds at room temperature, then **discard ethanol**.
9. **Repeat wash** again (step 7-8).
10. Spin down, place the tube on the magnetic separator, and **discard residual ethanol**.
11. **Air-dry the sample tube** at room temperature for **2 minutes** or until residual ethanol is completely evaporated.
12. **Add 32 µL of nuclease-free water** to the sample tube and vortex to mix. Pulse spin down and **incubate for 2 minutes** at room temperature.
13. Place the tube on the magnetic separator for **2 minutes** until the solution is clear.
14. Transfer the supernatant (30 µL) to a new tube.

STOP POINT:
If you are not immediately continuing to the next step, store the sample at -20°C.

NOTE

3. To avoid the bead precipitation, we do not recommend spinning down the CeleMag Clean-up bead too strongly.

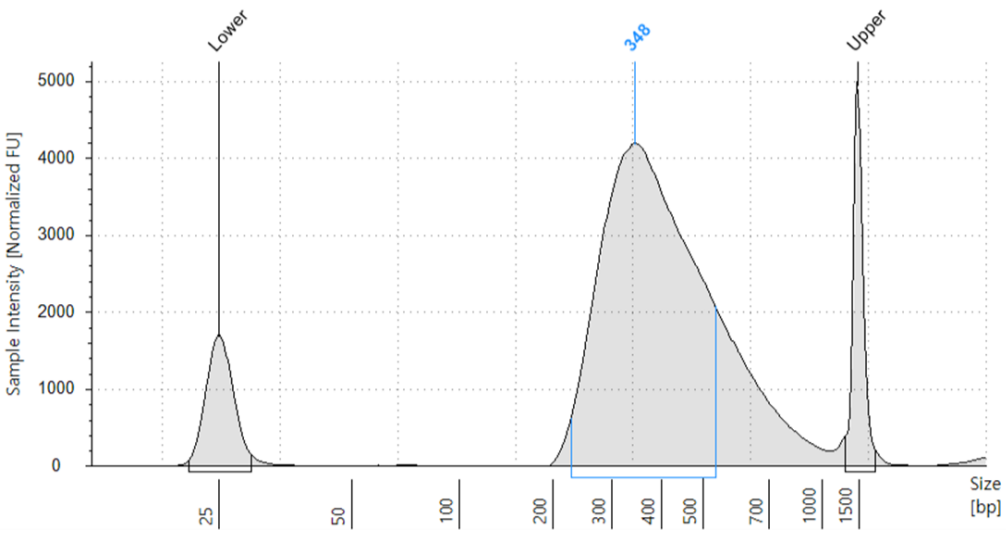
6. Do not disturb the bead pellet while you remove the supernatant.

11. Overdrying of the bead pellet will decrease the elution efficiency.

Step 6. Assess the Quality and Quantity of Each Index-Tagged Library

[Procedure]

1. Determine the quality and quantity of the sample with the appropriate instrument.



This figure shows the electrophoresis diagram detected by TapeStation.

2. Make sure that the size of the index-tagged library shows the main peak approximately 300-550 bp.

NOTE

Q1. What to do when "Bubble effect" appears on TapeStation or Bioanalyzer result?

A1. Measuring index-tagged library using a fluorometric method and using the expected peak size for quantity calculation is recommended.

Q2. Why does "Bubble effect" appears in some cases?

A2. Bubbled product has partially open conformation which makes slow migration in gel matrix based analyzer. Slower migration generates an artificial peak with bigger the size in base pairs of the expected library peak.

Q3. Can Index-tagged sample containing bubbled product be used for sequencing?

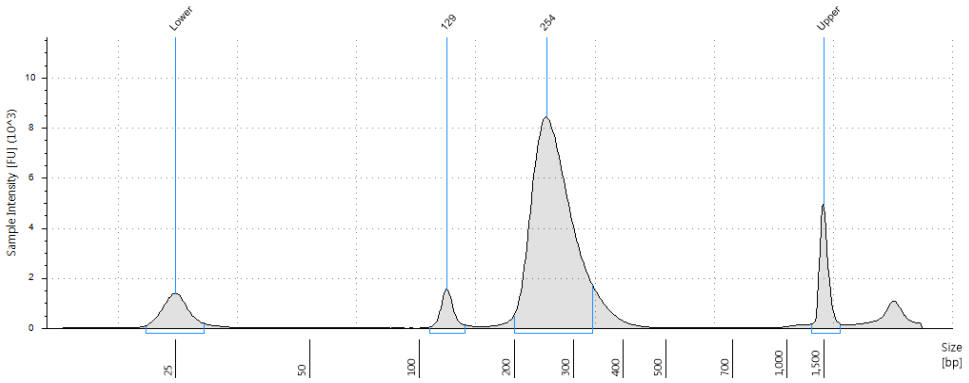
A3. They can be used for sequencing or hybridization target enrichment process. Bubble products will be denatured into single strands prior to flow cell binding or target enrichment probe binding, and single stranded DNA will bind to flow cell or target enrichment probes just as non-bubbled products.

Step 7. (Optional) Removal of Adapter Dimer

Adapter dimer in the pre-capture amplified library can be detected by gel-electrophoresis-based instruments such as TapeStation and Bioanalyzer.

[Procedure]

1. Remove the adapter dimer if the concentration of the adapter dimer is greater than 3.5 ng/µL.



The electrophoresis diagram shows the adapter dimer at 129 bp.

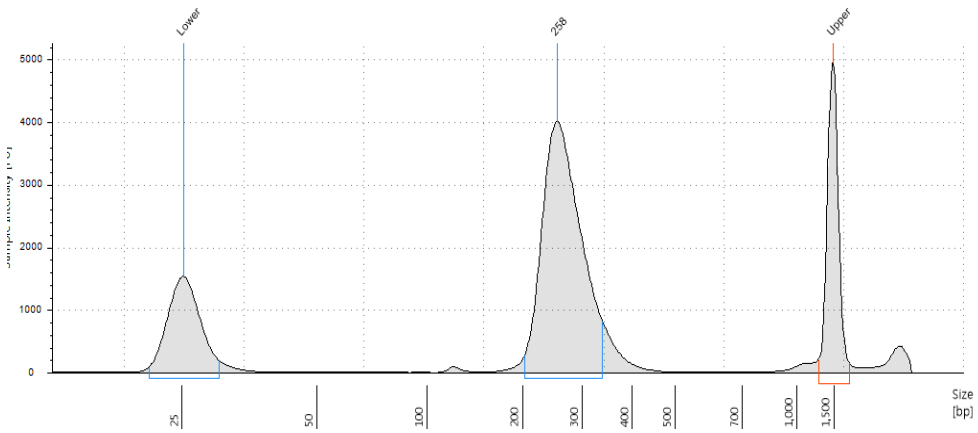
2. Purify the sample using the CeleMag Clean-up Bead based on the following instruction.

The Volume Ratio of CeleMag Clean-up Bead

Volume Ratio
Bead : Sample = 0.8 : 1

3. Follow the procedure in **'Step. Purification of Sample Using Clean-up Bead.'**

4. Confirm the removal of the adapter dimer through available instrumentation.



The electrophoresis diagram shows the removal of the adapter dimer at 129 bp.

NOTE

Appendix: Optimize Celemics Kit Protocol Based on Your Samples

[Note]

Celemics EP-kit is sensitive to several factors, including the amount of input DNA, DNA quality, fragmentation time, PCR cycle, and other experimental conditions. This appendix will provide several recommendations based on our experience for successful library preparation.

A. EDTA Containing Samples

In some cases, samples can be stored in buffers containing EDTA. The presence of EDTA is critical to the efficiency of Celemics EP-kit performance. Thus, we recommend you clean-up the sample to remove EDTA.

B. Fragmentation Time

Celemics EP-kit has a DNA fragmentation step in "NGS Library Preparation, Step: Enzymatic Preparation." In this step, fragmentation time could be determined by the amount/quality/origin of input DNA.

The following table suggests fragmentation time based on the amount of input gDNA. (For FFPE samples, see appendix D)

Recommendation for Fragmentation Time Based on the Amount of Input gDNA

gDNA Amount	Fragmentation Time
Amount ≤ 100 ng	The fragmentation times differ between enzyme production lots. Please find the fragmentation time-size table on the EP kit webpage*.
Amount > 100 ng	Slightly increase fragmentation time

* We do not recommend performing fragmentation with input gDNA less than 50ng. But if needed, increasing fragmentation time is suggested.

NOTE

* https://celemics.com/en/sub/search/CoA_of_Celemics_EP_Kit.asp

C. PCR cycles

Consider the number of PCR cycles if the sample is fragmented into target size. Having more cycles than needed may lead to undesired, overamplified 'bubble products' that show an additional peak in the electrophoresis diagram.

The following table provides suggestions based on our experimental data using high-quality reference genomic DNA(NA12878).

The Number of PCR Cycles for each Input gDNA Amount

gDNA Amount	Total PCR Cycles
50 ng	6-8
100 ng	5-7
200 ng	4-6

* Quality of gDNA may affect the result. If you are still trying to get enough DNA from our recommended PCR cycle, adjust PCR cycles to obtain a better result.

D. FFPE Samples

Celemics EP-kit also applies to FFPE samples but needs to be optimized. From our experiments, we suggest fragmentation time and PCR cycles based on DNA Integrity Number (DIN) score conditions in the following table.

Optimization Based on DIN Score

DIN	Fragmentation Time	PCR Cycles
3 ≤ DIN < 5	5-10 minutes less	Additional 1 or 2 cycles
DIN < 3	Do not highly depend on DIN	Additional 2 or more cycles

* For samples with DIN≥5, follow the protocol on the previous page, and see Appendix B and C to optimize fragmentation time and PCR cycles.
** The instruction does not guarantee library preparation outcomes when using low-quality samples with DIN less than 3.

NOTE

Appendix: UDI Index Sequences

UDI Index Sequences (1 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 1	ATATGCGC	GCGCATAT	CTGATCGT	UDI Primer 34	GATGTGTG	CACACATC	CATGGCTA
UDI Primer 2	TGGTACAG	CTGTACCA	ACTCTCGA	UDI Primer 35	CGCAATCT	AGATTGCG	ATGCCTGT
UDI Primer 3	AACCGTTC	GAACGGTT	TGAGCTAG	UDI Primer 36	TGGTAGCT	AGCTACCA	CAACACCT
UDI Primer 4	TAACCGGT	ACCGGTTA	GAGACGAT	UDI Primer 37	GATAGGCT	AGCCTATC	TGTGACTG
UDI Primer 5	GAACATCG	CGATGTTC	CTTGTCGA	UDI Primer 38	AGTGGATC	GATCCACT	GTCATCGA
UDI Primer 6	CCTTGTAG	CTACAAGG	TTCCAAGG	UDI Primer 39	TTGGACGT	ACGTCCAA	AGCACTTC
UDI Primer 7	TCAGGCTT	AAGCCTGA	CGCATGAT	UDI Primer 40	ATGACGTC	GACGTCAT	GAAGGAAG
UDI Primer 8	GTTCTCGT	ACGAGAAC	ACGGAACA	UDI Primer 41	GAAGTTGG	CCAACTTC	GTTGTTCG
UDI Primer 9	AGAACGAG	CTCGTTCT	CGGCTAAT	UDI Primer 42	CATACCAC	GTGGTATG	CGGTTGTT
UDI Primer 10	TGCTTCCA	TGGAAGCA	ATCGATCG	UDI Primer 43	CTGTTGAC	GTCAACAG	ACTGAGGT
UDI Primer 11	CTTCGACT	AGTCGAAG	GCAAGATC	UDI Primer 44	TGGCATGT	ACATGCCA	TGAAGACG
UDI Primer 12	CACCTGTT	AACAGGTG	GCTATCCT	UDI Primer 45	ATCGCCAT	ATGGCGAT	GTTACGCA
UDI Primer 13	ATCACACG	CGTGTGAT	TACGCTAC	UDI Primer 46	TTGCGAAG	CTTCGCAA	AGCGTGTT
UDI Primer 14	CCGTAAGA	TCTTACGG	TGGACTCT	UDI Primer 47	AGTTCGTC	GACGAACT	GATCGAGT
UDI Primer 15	TACGCCTT	AAGGCGTA	AGAGTAGC	UDI Primer 48	GAGCAGTA	TACTGCTC	ACAGCTCA
UDI Primer 16	CGACGTTA	TAACGTCG	ATCCAGAG	UDI Primer 49	ACAGCTCA	TGAGCTGT	GAGCAGTA
UDI Primer 17	ATGCACGA	TCGTGCAT	GACGATCT	UDI Primer 50	GATCGAGT	ACTCGATC	AGTTCGTC
UDI Primer 18	CCTGATTG	CAATCAGG	AACTGAGC	UDI Primer 51	AGCGTGTT	AACACGCT	TTGCGAAG
UDI Primer 19	GTAGGAGT	ACTCCTAC	CTTAGGAC	UDI Primer 52	GTTACGCA	TGCGTAAC	ATCGCCAT
UDI Primer 20	ACTAGGAG	CTCCTAGT	GTGCCATA	UDI Primer 53	TGAAGACG	CGTCTTCA	TGGCATGT
UDI Primer 21	CACTAGCT	AGCTAGTG	GAATCCGA	UDI Primer 54	ACTGAGGT	ACCTCAGT	CTGTTGAC
UDI Primer 22	ACGACTTG	CAAGTCGT	TCGCTGTT	UDI Primer 55	CGGTTGTT	AACAACCG	CATACCAC
UDI Primer 23	CGTGTGTA	TACACACG	TTCGTTGG	UDI Primer 56	GTTGTTCG	CGAACAAC	GAAGTTGG
UDI Primer 24	GTTGACCT	AGGTCAAC	AAGCACTG	UDI Primer 57	GAAGGAAG	CTTCCTTC	ATGACGTC
UDI Primer 25	ACTCCATC	GATGGAGT	CCTTGATC	UDI Primer 58	AGCACTTC	GAAGTGCT	TTGGACGT
UDI Primer 26	CAATGTGG	CCACATTG	GTCGAAGA	UDI Primer 59	GTCATCGA	TCGATGAC	AGTGGATC
UDI Primer 27	TTGCAGAC	GTCTGCAA	ACCACGAT	UDI Primer 60	TGTGACTG	CAGTCACA	GATAGGCT
UDI Primer 28	CAGTCCAA	TTGGACTG	GATTACCG	UDI Primer 61	CAACACCT	AGGTGTTG	TGGTAGCT
UDI Primer 29	ACGTTCAG	CTGAACGT	GCACAACT	UDI Primer 62	ATGCCTGT	ACAGGCAT	CGCAATCT
UDI Primer 30	AACGTCTG	CAGACGTT	GCGTCATT	UDI Primer 63	CATGGCTA	TAGCCATG	GATGTGTG
UDI Primer 31	TATCGGTC	GACCGATA	ATCCGGTA	UDI Primer 64	GTGAAGTG	CACTTCAC	GATTGCTC
UDI Primer 32	CGCTCTAT	ATAGAGCG	CGTTGCAA	UDI Primer 65	CGTTGCAA	TTGCAACG	CGCTCTAT
UDI Primer 33	GATTGCTC	GAGCAATC	GTGAAGTG	UDI Primer 66	ATCCGGTA	TACCGGAT	TATCGGTC

a3.0.0-b3.2.0-b3.1.0-c3.2.0-b3.1.0

UDI Index Sequences (2 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 67	GCGTCATT	AATGACGC	AACGTCTG	UDI Primer 101	CAGAATCG	CGATTCTG	CAAGGATC
UDI Primer 68	GCACAACT	AGTTGTGC	ACGTTCAG	UDI Primer 102	TGCTATCC	GGATAGCA	CCAACAGT
UDI Primer 69	GATTACCG	CGGTAATC	CAGTCCAA	UDI Primer 103	CTCCGTAT	ATACGGAG	TCACCTGA
UDI Primer 70	ACCACGAT	ATCGTGGT	TTGCAGAC	UDI Primer 104	GAGTAGAC	GTCTACTC	CGGAACTT
UDI Primer 71	GTCGAAGA	TCTTCGAC	CAATGTGG	UDI Primer 105	AGCCAATC	GATTGGCT	TACTGCTG
UDI Primer 72	CCTTGATC	GATCAAGG	ACTCCATC	UDI Primer 106	GCCAAGTA	TACTTGGC	TCAGACAG
UDI Primer 73	AAGCACTG	CAGTGCTT	GTTGACCT	UDI Primer 107	GTGCACAT	ATGTGCAC	GTAATGCC
UDI Primer 74	TTCGTTGG	CCAACGAA	CGTGTGTA	UDI Primer 108	AGCAAGTG	CACTTGCT	CCAAGCAT
UDI Primer 75	TCGCTGTT	AACAGCGA	ACGACTTG	UDI Primer 109	CGAGGAAT	ATTCCTCG	GCTACTTC
UDI Primer 76	GAATCCGA	TCGGATTC	CACTAGCT	UDI Primer 110	GTATGGCA	TGCCATAC	TGACTTGC
UDI Primer 77	GTGCCATA	TATGGCAC	ACTAGGAG	UDI Primer 111	AGGTCTGA	TCAGACCT	TCGCACTA
UDI Primer 78	CTTAGGAC	GTCCTAAG	GTAGGAGT	UDI Primer 112	CTCATTGC	GCAATGAG	CCTATAGG
UDI Primer 79	AACTGAGC	GCTCAGTT	CCTGATTG	UDI Primer 113	ACATGAGG	CCTCATGT	CCGTGAAT
UDI Primer 80	GACGATCT	AGATCGTC	ATGCACGA	UDI Primer 114	TGTCGTAG	CTACGACA	GTCACCTT
UDI Primer 81	ATCCAGAG	CTCTGGAT	CGACGTTA	UDI Primer 115	CGTCTTCA	TGAAGACG	GGATGCTA
UDI Primer 82	AGAGTAGC	GCTACTCT	TACGCCTT	UDI Primer 116	GTCGCTAA	TTAGCGAC	TTACCGTG
UDI Primer 83	TGGACTCT	AGAGTCCA	CCGTAAGA	UDI Primer 117	TCCTGGTA	TACCAGGA	TGGTCAGT
UDI Primer 84	TACGCTAC	GTAGCGTA	ATCACACG	UDI Primer 118	CGTCACTA	TAGTGACG	CTTGAAGG
UDI Primer 85	GCTATCCT	AGGATAGC	CACCTGTT	UDI Primer 119	TGACGTCT	AGACGTCA	CGATTAGG
UDI Primer 86	GCAAGATC	GATCTTGC	CTTCGACT	UDI Primer 120	CATGCGAA	TTCGCATG	AACGCAGT
UDI Primer 87	ATCGATCG	CGATCGAT	TGCTTCCA	UDI Primer 121	AAGCTCCT	AGGAGCTT	GACAACTC
UDI Primer 88	CGGCTAAT	ATTAGCCG	AGAACGAG	UDI Primer 122	AGGTAACG	CGTTACCT	CTTGAGCA
UDI Primer 89	ACGGAACA	TGTTCCGT	GTTCTCGT	UDI Primer 123	CAAGCTTG	CAAGCTTG	CCATCAAG
UDI Primer 90	CGCATGAT	ATCATGCG	TCAGGCTT	UDI Primer 124	TCCTCATG	CATGAGGA	AGTCATCG
UDI Primer 91	TTCCAAGG	CCTTGGAA	CCTTGTAG	UDI Primer 125	CACCGATA	TATCGGTG	TGAACCTG
UDI Primer 92	CTTGTCGA	TCGACAAG	GAACATCG	UDI Primer 126	TGTAAGCG	CGCTTACA	TATCCTGC
UDI Primer 93	GAGACGAT	ATCGTCTC	TAACCGGT	UDI Primer 127	GAACGGAT	ATCCGTTC	ACCTAGTC
UDI Primer 94	TGAGCTAG	CTAGCTCA	AACCGTTC	UDI Primer 128	GACCTAGA	TCTAGGTC	TAGGCTGA
UDI Primer 95	ACTCTCGA	TCGAGAGT	TGGTACAG	UDI Primer 129	ATCGGTGT	ACACCGAT	CATTAGGC
UDI Primer 96	CTGATCGT	ACGATCAG	ATATGCGC	UDI Primer 130	TGAGACCA	TGGTCTCA	GCTGTTGA
UDI Primer 97	ATCTGCCT	AGGCAGAT	AGACTGTC	UDI Primer 131	ACGACACT	AGTGTCGT	TTCACAGC
UDI Primer 98	ATTCTGCG	CGCAGAAT	CAGGTTCA	UDI Primer 132	CGGACTAA	TTAGTCCG	ATGGTCAC
UDI Primer 99	ATCAAGCC	GGCTTGAT	AACTCGTG	UDI Primer 133	TAGCATCC	GGATGCTA	TCATTCGG
UDI Primer 100	GACTGACT	AGTCAGTC	GCAGATTC	UDI Primer 134	AAGTTGGC	GCCAACTT	AGGCGTAA

UDI Index Sequences (3 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 135	GATCCTCA	TGAGGATC	AGATACGG	UDI Primer 169	AACGCAGT	ACTGCGTT	CATGCGAA
UDI Primer 136	GTGACCAA	TTGGTCAC	AATGGCCT	UDI Primer 170	CGATTAGG	CCTAATCG	TGACGTCT
UDI Primer 137	CAGATGAG	CTCATCTG	CGACAAGT	UDI Primer 171	CTTGAAGG	CCTTCAAG	CGTCACTA
UDI Primer 138	TAACGACC	GGTCGTTA	ACACGTAC	UDI Primer 172	TGGTCAGT	ACTGACCA	TCCTGGTA
UDI Primer 139	CCAGTGAT	ATCACTGG	AACAGTCG	UDI Primer 173	TTACCGTG	CACGGTAA	GTCGCTAA
UDI Primer 140	GCTCAAGA	TCTTGAGC	TACATGCG	UDI Primer 174	GGATGCTA	TAGCATCC	CGTCTTCA
UDI Primer 141	AGAGAGCT	AGCTCTCT	GCATCACA	UDI Primer 175	GTCACCTT	AAGGTGAC	TGTCGTAG
UDI Primer 142	AGATCCAC	GTGGATCT	CTTATCCG	UDI Primer 176	CCGTGAAT	ATTCACGG	ACATGAGG
UDI Primer 143	CCTGACAA	TTGTCAGG	CGTAAGGT	UDI Primer 177	CCTATAGG	CCTATAGG	CTCATTGC
UDI Primer 144	CCGTATCT	AGATACGG	GACGTGAT	UDI Primer 178	TCGCACTA	TAGTGCGA	AGGTCTGA
UDI Primer 145	GACGTGAT	ATCACGTC	CCGTATCT	UDI Primer 179	TGACTTGC	GCAAGTCA	GTATGGCA
UDI Primer 146	CGTAAGGT	ACCTTACG	CCTGACAA	UDI Primer 180	GCTACTTC	GAAGTAGC	CGAGGAAT
UDI Primer 147	CTTATCCG	CGGATAAG	AGATCCAC	UDI Primer 181	CCAAGCAT	ATGCTTGG	AGCAAGTG
UDI Primer 148	GCATCACA	TGTGATGC	AGAGAGCT	UDI Primer 182	GTAATGCC	GGCATTAC	GTGCACAT
UDI Primer 149	TACATGCG	CGCATGTA	GCTCAAGA	UDI Primer 183	TCAGACAG	CTGTCTGA	GCCAAGTA
UDI Primer 150	AACAGTCG	CGACTGTT	CCAGTGAT	UDI Primer 184	TACTGCTG	CAGCAGTA	AGCCAATC
UDI Primer 151	ACACGTAC	GTACGTGT	TAACGACC	UDI Primer 185	CGGAACTT	AAGTTCCG	GAGTAGAC
UDI Primer 152	CGACAAGT	ACTTGTCG	CAGATGAG	UDI Primer 186	TCACCTGA	TCAGGTGA	CTCCGTAT
UDI Primer 153	AATGGCCT	AGGCCATT	GTGACCAA	UDI Primer 187	CCAACAGT	ACTGTTGG	TGCTATCC
UDI Primer 154	AGATACGG	CCGTATCT	GATCCTCA	UDI Primer 188	CAAGGATC	GATCCTTG	CAGAATCG
UDI Primer 155	AGGCGTAA	TTACGCCT	AAGTTGGC	UDI Primer 189	GCAGATTC	GAATCTGC	GACTGACT
UDI Primer 156	TCATTCGG	CCGAATGA	TAGCATCC	UDI Primer 190	AACTCGTG	CACGAGTT	ATCAAGCC
UDI Primer 157	ATGGTCAC	GTGACCAT	CGGACTAA	UDI Primer 191	CAGGTTCA	TGAACCTG	ATTCTGCG
UDI Primer 158	TTCACAGC	GCTGTGAA	ACGACACT	UDI Primer 192	AGACTGTC	GACAGTCT	ATCTGCCT
UDI Primer 159	GCTGTTGA	TCAACAGC	TGAGACCA	UDI Primer 193	TCCAACGT	ACGTTGGA	CCATATGC
UDI Primer 160	CATTAGGC	GCCTAATG	ATCGGTGT	UDI Primer 194	ACTATGCC	GGCATAGT	TTGTGACC
UDI Primer 161	TAGGCTGA	TCAGCCTA	GACCTAGA	UDI Primer 195	TAGTCGAG	CTCGACTA	ACAGGTTG
UDI Primer 162	ACCTAGTC	GACTAGGT	GAACGGAT	UDI Primer 196	TAACGCAG	CTGCGTTA	TGCGCAAT
UDI Primer 163	TATCCTGC	GCAGGATA	TGTAAGCG	UDI Primer 197	TCGGTCAA	TTGACCGA	TTCAGGAG
UDI Primer 164	TGAACCTG	CAGGTTCA	CACCGATA	UDI Primer 198	GGTATCTG	CAGATACC	TACTCTGG
UDI Primer 165	AGTCATCG	CGATGACT	TCCTCATG	UDI Primer 199	CTCTCAGA	TCTGAGAG	ACCTTCGT
UDI Primer 166	CCATCAAG	CTTGATGG	CAAGCTTG	UDI Primer 200	TGCAGTTG	CAACTGCA	CAGGCTAT
UDI Primer 167	CTTGAGCA	TGCTCAAG	AGGTAACG	UDI Primer 201	CAATTGCG	CGCAATTG	TCGTATGG
UDI Primer 168	GACAACTC	GAGTTGTC	AAGCTCCT	UDI Primer 202	TCTACCAG	CTGGTAGA	TGCGCTTA

UDI Index Sequences (4 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 203	CCTGTTAC	GTAACAGG	TGAAGGAC	UDI Primer 237	GTCAATGG	CCATTGAC	AATGACGG
UDI Primer 204	AAGGCGTT	AACGCCTT	ACGGACAT	UDI Primer 238	GTCAGACA	TGTCTGAC	CCACAGTA
UDI Primer 205	TACCTGAC	GTCAGGTA	CTAAGCCA	UDI Primer 239	CTCTGATG	CATCAGAG	GTCTTCAG
UDI Primer 206	ATTGCACC	GGTGCAAT	TCTAGTCG	UDI Primer 240	CTCCATTG	CAATGGAG	TGTGCAGA
UDI Primer 207	TCAGTTCG	CGAACTGA	TCTAAGGC	UDI Primer 241	TGTGCAGA	TCTGCACA	CTCCATTG
UDI Primer 208	AGATGACC	GGTCATCT	CACACTGT	UDI Primer 242	GTCTTCAG	CTGAAGAC	CTCTGATG
UDI Primer 209	TTAGCTGC	GCAGCTAA	ACAAGGCT	UDI Primer 243	CCACAGTA	TACTGTGG	GTCAGACA
UDI Primer 210	GGACAGTT	AACTGTCC	TTGTCCGA	UDI Primer 244	AATGACGG	CCGTCATT	GTCAATGG
UDI Primer 211	AATCGCAC	GTGCGATT	GCATTCTC	UDI Primer 245	CACTGTCA	TGACAGTG	TGAGGTTC
UDI Primer 212	CAAGGTCT	AGACCTTG	TAGTCCTC	UDI Primer 246	TGCCTAGT	ACTAGGCA	GTTGAGAC
UDI Primer 213	ATGGCTAG	CTAGCCAT	TCTCGGAT	UDI Primer 247	CGGTATTG	CAATACCG	GGAACACT
UDI Primer 214	ACGGTTCT	AGAACCGT	GCGAATGT	UDI Primer 248	CAGATAGC	GCTATCTG	CACACGTA
UDI Primer 215	CCAGTCTA	TAGACTGG	GTGATCTC	UDI Primer 249	ATCGTCCA	TGGACGAT	AGTGCGAT
UDI Primer 216	AGCATGGA	TCCATGCT	TTCCTGGA	UDI Primer 250	CGTAGAGA	TCTCTACG	ACCATTGG
UDI Primer 217	CAGGCATA	TATGCCTG	CCATGGTT	UDI Primer 251	GGTCGATA	TATCGACC	GAGCTTGT
UDI Primer 218	CATAGACC	GGTCTATG	ACGAGTGA	UDI Primer 252	GCAGTAAC	GTTACTGC	GTGAGTCT
UDI Primer 219	AAGCAGGT	ACCTGCTT	GGCTAACA	UDI Primer 253	TGTCCGTA	TACGGACA	GGACCTAT
UDI Primer 220	AGTAACGC	GCGTTACT	TGACTGCA	UDI Primer 254	CTAGCACT	AGTGCTAG	TCTTGCCA
UDI Primer 221	CATCTACG	CGTAGATG	TGATCGGA	UDI Primer 255	ACTAGAGC	GCTCTAGT	CGAATTCC
UDI Primer 222	TAGAGCAC	GTGCTCTA	GGCTTGAA	UDI Primer 256	AGCCACAT	ATGTGGCT	ATGCTTCC
UDI Primer 223	TTCGAACC	GGTTCGAA	TGTGATGC	UDI Primer 257	TCGTTACG	CGTAACGA	GAGTCTTG
UDI Primer 224	GAGTCTTG	CAAGACTC	TCGTTACG	UDI Primer 258	TGTGATGC	GCATCACA	TTCGAACC
UDI Primer 225	ATGCTTCC	GGAAGCAT	AGCCACAT	UDI Primer 259	GGCTTGAA	TTCAAGCC	TAGAGCAC
UDI Primer 226	CGAATTCC	GGAATTCG	ACTAGAGC	UDI Primer 260	TGATCGGA	TCCGATCA	CATCTACG
UDI Primer 227	TCTTGCCA	TGGCAAGA	CTAGCACT	UDI Primer 261	TGACTGCA	TGCAGTCA	AGTAACGC
UDI Primer 228	GGACCTAT	ATAGGTCC	TGTCCGTA	UDI Primer 262	GGCTAACA	TGTTAGCC	AAGCAGGT
UDI Primer 229	GTGAGTCT	AGACTCAC	GCAGTAAC	UDI Primer 263	ACGAGTGA	TCACTCGT	CATAGACC
UDI Primer 230	GAGCTTGT	ACAAGCTC	GGTCGATA	UDI Primer 264	CCATGGTT	AACCATGG	CAGGCATA
UDI Primer 231	ACCATTGG	CCAATGGT	CGTAGAGA	UDI Primer 265	TTCCTGGA	TCCAGGAA	AGCATGGA
UDI Primer 232	AGTGCGAT	ATCGCACT	ATCGTCCA	UDI Primer 266	GTGATCTC	GAGATCAC	CCAGTCTA
UDI Primer 233	CACACGTA	TACGTGTG	CAGATAGC	UDI Primer 267	GCGAATGT	ACATTCGC	ACGGTTCT
UDI Primer 234	GGAACACT	AGTGTTCC	CGGTATTG	UDI Primer 268	TCTCGGAT	ATCCGAGA	ATGGCTAG
UDI Primer 235	GTTGAGAC	GTCTCAAC	TGCCTAGT	UDI Primer 269	TAGTCCTC	GAGGACTA	CAAGGTCT
UDI Primer 236	TGAGGTTC	GAACCTCA	CACTGTCA	UDI Primer 270	GCATTCTC	GAGAATGC	AATCGCAC

UDI Index Sequences (5 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 271	TTGTCCGA	TCGGACAA	GGACAGTT	UDI Primer 305	CTTCACAG	CTGTGAAG	CTCGATAC
UDI Primer 272	ACAAGGCT	AGCCTTGT	TTAGCTGC	UDI Primer 306	TCGAGGAA	TTCCTCGA	AGCCTAAG
UDI Primer 273	CACACTGT	ACAGTGTG	AGATGACC	UDI Primer 307	GTTGCATG	CATGCAAC	GCATTGGA
UDI Primer 274	TCTAAGGC	GCCTTAGA	TCAGTTCG	UDI Primer 308	GGTACCAT	ATGGTACC	CAATCGAC
UDI Primer 275	TCTAGTCG	CGACTAGA	ATTGCACC	UDI Primer 309	ATACCAGC	GCTGGTAT	TTGGCACA
UDI Primer 276	CTAAGCCA	TGGCTTAG	TACCTGAC	UDI Primer 310	GTCTCGTA	TACGAGAC	CCGAATAC
UDI Primer 277	ACGGACAT	ATGTCCGT	AAGGCGTT	UDI Primer 311	TCCGTATC	GATACGGA	ATGGTACG
UDI Primer 278	TGAAGGAC	GTCCTTCA	CCTGTTAC	UDI Primer 312	GTTCGAAC	GTTCGAAC	GGCTTATG
UDI Primer 279	TGCGCTTA	TAAGCGCA	TCTACCAG	UDI Primer 313	TTCTAGCG	CGCTAGAA	TTGCGTCA
UDI Primer 280	TCGTATGG	CCATACGA	CAATTGCG	UDI Primer 314	TTACGCGA	TCGCGTAA	ATGGAAGC
UDI Primer 281	CAGGCTAT	ATAGCCTG	TGCAGTTG	UDI Primer 315	GTCCGATT	AATCGGAC	GCTTAGCT
UDI Primer 282	ACCTTCGT	ACGAAGGT	CTCTCAGA	UDI Primer 316	CATAGGTG	CACCTATG	TACATCGC
UDI Primer 283	TACTCTGG	CCAGAGTA	GGTATCTG	UDI Primer 317	TAGACGCA	TGCGTCTA	TGCTGAAC
UDI Primer 284	TTCAGGAG	CTCCTGAA	TCGGTCAA	UDI Primer 318	GACTGTAG	CTACAGTC	ATCGGAAG
UDI Primer 285	TGCGCAAT	ATTGCGCA	TAACGCAG	UDI Primer 319	GACGAATG	CATTCGTC	GGCATTAG
UDI Primer 286	ACAGGTTG	CAACCTGT	TAGTCGAG	UDI Primer 320	AGTGGTCA	TGACCACT	GAACTCAC
UDI Primer 287	TTGTGACC	GGTCACAA	ACTATGCC	UDI Primer 321	CTCTGGAA	TTCCAGAG	ATTGGCTG
UDI Primer 288	CCATATGC	GCATATGG	TCCAACGT	UDI Primer 322	TCTGGAAC	GTTCCAGA	CTAGCGTA
UDI Primer 289	AACGTTCC	GGAACGTT	CCTCTGAA	UDI Primer 323	CAGCTTAC	GTAAGCTG	ATAGGTCC
UDI Primer 290	CCACTTAG	CTAAGTGG	AGCATCCT	UDI Primer 324	TGTGGCAT	ATGCCACA	GGTTAAGC
UDI Primer 291	TCGCCAAT	ATTGGCGA	GGATTGCT	UDI Primer 325	GGTCATTC	GAATGACC	CGAAGTGT
UDI Primer 292	TGCAGCAA	TTGCTGCA	CGCATATC	UDI Primer 326	TACGAGCA	TGCTCGTA	TAAGTGGC
UDI Primer 293	CGAATGTG	CACATTCG	GTAACGGT	UDI Primer 327	ACTGACTC	GAGTCAGT	CCGTGTTA
UDI Primer 294	CACAGCTT	AAGCTGTG	TCGTCTCA	UDI Primer 328	CTCGAGTT	AACTCGAG	TTCTCGAC
UDI Primer 295	GCATGTGT	ACACATGC	GGTTCTAG	UDI Primer 329	GTATCAGG	CCTGATAC	ACGTCGAA
UDI Primer 296	CTAGACTG	CAGTCTAG	CTTAGTGG	UDI Primer 330	GGAACGTA	TACGTTCC	AATGGTGC
UDI Primer 297	TCGGAATG	CATTCCGA	ATACCGCA	UDI Primer 331	CGAGTACA	TGTACTCG	TGGATCGA
UDI Primer 298	CACTAAGG	CCTTAGTG	CTGCTTGA	UDI Primer 332	ACTTCACG	CGTGAAGT	GTGGATCA
UDI Primer 299	GCTAGCAA	TTGCTAGC	ATGCGATC	UDI Primer 333	GCTGGTAT	ATACCAGC	TGTCACAC
UDI Primer 300	ATGAACCG	CGGTTCAT	CGATAGTC	UDI Primer 334	CACGACAT	ATGTCGTG	TGTGTACC
UDI Primer 301	TTCTACGC	GCGTAGAA	CAGGTCTT	UDI Primer 335	TGAGTCAC	GTGACTCA	GTACCTTC
UDI Primer 302	TGGCGATT	AATCGCCA	TGGACATC	UDI Primer 336	AGAGCCTA	TAGGCTCT	ATTGCGGA
UDI Primer 303	GTACTCCA	TGGAGTAC	AAGAGCGT	UDI Primer 337	ATTGCGGA	TCCGCAAT	AGAGCCTA
UDI Primer 304	AAGTGTCC	GGACACTT	ACGCATTC	UDI Primer 338	GTACCTTC	GAAGGTAC	TGAGTCAC

UDI Index Sequences (6 of 6)

Index	i5		i7	Index	i5		i7
	Forward	Reverse			Forward	Reverse	
UDI Primer 339	TGTGTACC	GGTACACA	CACGACAT	UDI Primer 372	CAGGTCTT	AAGACCTG	TTCTACGC
UDI Primer 340	TGTCACAC	GTGTGACA	GCTGGTAT	UDI Primer 373	CGATAGTC	GACTATCG	ATGAACCG
UDI Primer 341	GTGGATCA	TGATCCAC	ACTTCACG	UDI Primer 374	ATGCGATC	GATCGCAT	GCTAGCAA
UDI Primer 342	TGGATCGA	TCGATCCA	CGAGTACA	UDI Primer 375	CTGCTTGA	TCAAGCAG	CACTAAGG
UDI Primer 343	AATGGTGC	GCACCATT	GGAACGTA	UDI Primer 376	ATACCGCA	TGCGGTAT	TCGGAATG
UDI Primer 344	ACGTCGAA	TTCGACGT	GTATCAGG	UDI Primer 377	CTTAGTGG	CCACTAAG	CTAGACTG
UDI Primer 345	TTCTCGAC	GTCGAGAA	CTCGAGTT	UDI Primer 378	GGTTCTAG	CTAGAACC	GCATGTGT
UDI Primer 346	CCGTGTTA	TAACACGG	ACTGACTC	UDI Primer 379	TCGTCTCA	TGAGACGA	CACAGCTT
UDI Primer 347	TAAGTGGC	GCCACTTA	TACGAGCA	UDI Primer 380	GTAACGGT	ACCGTTAC	CGAATGTG
UDI Primer 348	CGAAGTGT	ACACTTCG	GGTCATTC	UDI Primer 381	CGCATATC	GATATGCG	TGCAGCAA
UDI Primer 349	GGTTAAGC	GCTTAACC	TGTGGCAT	UDI Primer 382	GGATTGCT	AGCAATCC	TCGCCAAT
UDI Primer 350	ATAGGTCC	GGACCTAT	CAGCTTAC	UDI Primer 383	AGCATCCT	AGGATGCT	CCACTTAG
UDI Primer 351	CTAGCGTA	TACGCTAG	TCTGGAAC	UDI Primer 384	CCTCTGAA	TTCAGAGG	AACGTTCC
UDI Primer 352	ATTGGCTG	CAGCCAAT	CTCTGGAA				
UDI Primer 353	GAACTCAC	GTGAGTTC	AGTGGTCA				
UDI Primer 354	GGCATTAG	CTAATGCC	GACGAATG				
UDI Primer 355	ATCGGAAG	CTTCCGAT	GACTGTAG				
UDI Primer 356	TGCTGAAC	GTTCAGCA	TAGACGCA				
UDI Primer 357	TACATCGC	GCGATGTA	CATAGGTG				
UDI Primer 358	GCTTAGCT	AGCTAAGC	GTCCGATT				
UDI Primer 359	ATGGAAGC	GCTTCCAT	TTACGCGA				
UDI Primer 360	TTGCGTCA	TGACGCAA	TTCTAGCG				
UDI Primer 361	GGCTTATG	CATAAGCC	GTTCGAAC				
UDI Primer 362	ATGGTACG	CGTACCAT	TCCGTATC				
UDI Primer 363	CCGAATAC	GTATTCGG	GTCTCGTA				
UDI Primer 364	TTGGCACA	TGTGCCAA	ATACCAGC				
UDI Primer 365	CAATCGAC	GTCGATTG	GGTACCAT				
UDI Primer 366	GCATTGGA	TCCAATGC	GTTGCATG				
UDI Primer 367	AGCCTAAG	CTTAGGCT	TCGAGGAA				
UDI Primer 368	CTCGATAC	GTATCGAG	CTTCACAG				
UDI Primer 369	ACGCATTC	GAATGCGT	AAGTGTCC				
UDI Primer 370	AAGAGCGT	ACGCTCTT	GTACTCCA				
UDI Primer 371	TGGACATC	GATGTCCA	TGGCGATT				

i5 Index Sequence Orientation *		
Forward	Reverse	
MiSeq	HiSeq3000	NextSeq 500
HiSeq2000	HiSeq4000	NextSeq 550
HiSeq2500	HiSeq X	NextSeq 1000
NovaSeq 6000 v1.0	NovaSeq 6000 v1.5	NextSeq 2000
MiniSeq Rapid	NovaSeq X	iSeq100
	NovaSeq X Plus	MiniSeq Standard

* Check the index sequence orientation based on the Sequencer in use.

Appendix: UDI Index Plate Layout

Refer to this appendix when using index plates. Check the table for index numbers and well positions according to the purchased unit size and index type.

For 16 Sample Kit

16-A Set

	1	2
A	U001	U009
B	U002	U010
C	U003	U011
D	U004	U012
E	U005	U013
F	U006	U014
G	U007	U015
H	U008	U016

16-B Set

	1	2
A	U017	U025
B	U018	U026
C	U019	U027
D	U020	U028
E	U021	U029
F	U022	U030
G	U023	U031
H	U024	U032

16-C Set

	1	2
A	U033	U041
B	U034	U042
C	U035	U043
D	U036	U044
E	U037	U045
F	U038	U046
G	U039	U047
H	U040	U048

16-D Set

	1	2
A	U049	U057
B	U050	U058
C	U051	U059
D	U052	U060
E	U053	U061
F	U054	U062
G	U055	U063
H	U056	U064

16-E Set

	1	2
A	U065	U073
B	U066	U074
C	U067	U075
D	U068	U076
E	U069	U077
F	U070	U078
G	U071	U079
H	U072	U080

16-F Set

	1	2
A	U081	U089
B	U082	U090
C	U083	U091
D	U084	U092
E	U085	U093
F	U086	U094
G	U087	U095
H	U088	U096

16-G Set

	1	2
A	U097	U105
B	U098	U106
C	U099	U107
D	U100	U108
E	U101	U109
F	U102	U110
G	U103	U111
H	U104	U112

16-H Set

	1	2
A	U113	U121
B	U114	U122
C	U115	U123
D	U116	U124
E	U117	U125
F	U118	U126
G	U119	U127
H	U120	U128

16-I Set

	1	2
A	U129	U137
B	U130	U138
C	U131	U139
D	U132	U140
E	U133	U141
F	U134	U142
G	U135	U143
H	U136	U144

16-J Set

	1	2
A	U145	U153
B	U146	U154
C	U147	U155
D	U148	U156
E	U149	U157
F	U150	U158
G	U151	U159
H	U152	U160

16-K Set

	1	2
A	U161	U169
B	U162	U170
C	U163	U171
D	U164	U172
E	U165	U173
F	U166	U174
G	U167	U175
H	U168	U176

16-L Set

	1	2
A	U177	U185
B	U178	U186
C	U179	U187
D	U180	U188
E	U181	U189
F	U182	U190
G	U183	U191
H	U184	U192

a3.0.0-b3.2.0-b3.1.0-c3.2.0-b3.1.0

16-M Set

	1	2
A	U193	U201
B	U194	U202
C	U195	U203
D	U196	U204
E	U197	U205
F	U198	U206
G	U199	U207
H	U200	U208

16-N Set

	1	2
A	U209	U217
B	U210	U218
C	U211	U219
D	U212	U220
E	U213	U221
F	U214	U222
G	U215	U223
H	U216	U224

16-O Set

	1	2
A	U225	U233
B	U226	U234
C	U227	U235
D	U228	U236
E	U229	U237
F	U230	U238
G	U231	U239
H	U232	U240

16-P Set

	1	2
A	U241	U249
B	U242	U250
C	U243	U251
D	U244	U252
E	U245	U253
F	U246	U254
G	U247	U255
H	U248	U256

16-Q Set

	1	2
A	U257	U265
B	U258	U266
C	U259	U267
D	U260	U268
E	U261	U269
F	U262	U270
G	U263	U271
H	U264	U272

16-R Set

	1	2
A	U273	U281
B	U274	U282
C	U275	U283
D	U276	U284
E	U277	U285
F	U278	U286
G	U279	U287
H	U280	U288

16-S Set

	1	2
A	U289	U297
B	U290	U298
C	U291	U299
D	U292	U300
E	U293	U301
F	U294	U302
G	U295	U303
H	U296	U304

16-T Set

	1	2
A	U305	U313
B	U306	U314
C	U307	U315
D	U308	U316
E	U309	U317
F	U310	U318
G	U311	U319
H	U312	U320

16-U Set

	1	2
A	U321	U329
B	U322	U330
C	U323	U331
D	U324	U332
E	U325	U333
F	U326	U334
G	U327	U335
H	U328	U336

16-V Set

	1	2
A	U337	U345
B	U338	U346
C	U339	U347
D	U340	U348
E	U341	U349
F	U342	U350
G	U343	U351
H	U344	U352

16-W Set

	1	2
A	U353	U361
B	U354	U362
C	U355	U363
D	U356	U364
E	U357	U365
F	U358	U366
G	U359	U367
H	U360	U368

16-X Set

	1	2
A	U369	U377
B	U370	U378
C	U371	U379
D	U372	U380
E	U373	U381
F	U374	U382
G	U375	U383
H	U376	U384

For 32 Sample Kit

32-A Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U001	U009	U017	U025								
B	U002	U010	U018	U026								
C	U003	U011	U019	U027								
D	U004	U012	U020	U028								
E	U005	U013	U021	U029								
F	U006	U014	U022	U030								
G	U007	U015	U023	U031								
H	U008	U016	U024	U032								

32-B Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U033	U041	U049	U057								
B	U034	U042	U050	U058								
C	U035	U043	U051	U059								
D	U036	U044	U052	U060								
E	U037	U045	U053	U061								
F	U038	U046	U054	U062								
G	U039	U047	U055	U063								
H	U040	U048	U056	U064								

32-C Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U065	U073	U081	U089								
B	U066	U074	U082	U090								
C	U067	U075	U083	U091								
D	U068	U076	U084	U092								
E	U069	U077	U085	U093								
F	U070	U078	U086	U094								
G	U071	U079	U087	U095								
H	U072	U080	U088	U096								

32-D Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U097	U105	U113	U121								
B	U098	U106	U114	U122								
C	U099	U107	U115	U123								
D	U100	U108	U116	U124								
E	U101	U109	U117	U125								
F	U102	U110	U118	U126								
G	U103	U111	U119	U127								
H	U104	U112	U120	U128								

32-E Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U129	U137	U145	U153								
B	U130	U138	U146	U154								
C	U131	U139	U147	U155								
D	U132	U140	U148	U156								
E	U133	U141	U149	U157								
F	U134	U142	U150	U158								
G	U135	U143	U151	U159								
H	U136	U144	U152	U160								

32-F Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U161	U169	U177	U185								
B	U162	U170	U178	U186								
C	U163	U171	U179	U187								
D	U164	U172	U180	U188								
E	U165	U173	U181	U189								
F	U166	U174	U182	U190								
G	U167	U175	U183	U191								
H	U168	U176	U184	U192								

32-G Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U193	U201	U209	U217								
B	U194	U202	U210	U218								
C	U195	U203	U211	U219								
D	U196	U204	U212	U220								
E	U197	U205	U213	U221								
F	U198	U206	U214	U222								
G	U199	U207	U215	U223								
H	U200	U208	U216	U224								

32-H Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U225	U233	U241	U249								
B	U226	U234	U242	U250								
C	U227	U235	U243	U251								
D	U228	U236	U244	U252								
E	U229	U237	U245	U253								
F	U230	U238	U246	U254								
G	U231	U239	U247	U255								
H	U232	U240	U248	U256								

32-I Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U257	U265	U273	U281								
B	U258	U266	U274	U282								
C	U259	U267	U275	U283								
D	U260	U268	U276	U284								
E	U261	U269	U277	U285								
F	U262	U270	U278	U286								
G	U263	U271	U279	U287								
H	U264	U272	U280	U288								

32-J Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U289	U297	U305	U313								
B	U290	U298	U306	U314								
C	U291	U299	U307	U315								
D	U292	U300	U308	U316								
E	U293	U301	U309	U317								
F	U294	U302	U310	U318								
G	U295	U303	U311	U319								
H	U296	U304	U312	U320								

32-K Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U321	U329	U337	U345								
B	U322	U330	U338	U346								
C	U323	U331	U339	U347								
D	U324	U332	U340	U348								
E	U325	U333	U341	U349								
F	U326	U334	U342	U350								
G	U327	U335	U343	U351								
H	U328	U336	U344	U352								

32-L Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U353	U361	U369	U377								
B	U354	U362	U370	U378								
C	U355	U363	U371	U379								
D	U356	U364	U372	U380								
E	U357	U365	U373	U381								
F	U358	U366	U374	U382								
G	U359	U367	U375	U383								
H	U360	U368	U376	U384								

For 48 Sample Kit

48-A Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U001	U009	U017	U025	U033	U041						
B	U002	U010	U018	U026	U034	U042						
C	U003	U011	U019	U027	U035	U043						
D	U004	U012	U020	U028	U036	U044						
E	U005	U013	U021	U029	U037	U045						
F	U006	U014	U022	U030	U038	U046						
G	U007	U015	U023	U031	U039	U047						
H	U008	U016	U024	U032	U040	U048						

48-B Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U049	U057	U065	U073	U081	U089						
B	U050	U058	U066	U074	U082	U090						
C	U051	U059	U067	U075	U083	U091						
D	U052	U060	U068	U076	U084	U092						
E	U053	U061	U069	U077	U085	U093						
F	U054	U062	U070	U078	U086	U094						
G	U055	U063	U071	U079	U087	U095						
H	U056	U064	U072	U080	U088	U096						

48-C Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U097	U105	U113	U121	U129	U137						
B	U098	U106	U114	U122	U130	U138						
C	U099	U107	U115	U123	U131	U139						
D	U100	U108	U116	U124	U132	U140						
E	U101	U109	U117	U125	U133	U141						
F	U102	U110	U118	U126	U134	U142						
G	U103	U111	U119	U127	U135	U143						
H	U104	U112	U120	U128	U136	U144						

48-D Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U145	U153	U161	U169	U177	U185						
B	U146	U154	U162	U170	U178	U186						
C	U147	U155	U163	U171	U179	U187						
D	U148	U156	U164	U172	U180	U188						
E	U149	U157	U165	U173	U181	U189						
F	U150	U158	U166	U174	U182	U190						
G	U151	U159	U167	U175	U183	U191						
H	U152	U160	U168	U176	U184	U192						

48-E Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U193	U201	U209	U217	U225	U233						
B	U194	U202	U210	U218	U226	U234						
C	U195	U203	U211	U219	U227	U235						
D	U196	U204	U212	U220	U228	U236						
E	U197	U205	U213	U221	U229	U237						
F	U198	U206	U214	U222	U230	U238						
G	U199	U207	U215	U223	U231	U239						
H	U200	U208	U216	U224	U232	U240						

48-F Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U241	U249	U257	U265	U273	U281						
B	U242	U250	U258	U266	U274	U282						
C	U243	U251	U259	U267	U275	U283						
D	U244	U252	U260	U268	U276	U284						
E	U245	U253	U261	U269	U277	U285						
F	U246	U254	U262	U270	U278	U286						
G	U247	U255	U263	U271	U279	U287						
H	U248	U256	U264	U272	U280	U288						

48-G Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U289	U297	U305	U313	U321	U329						
B	U290	U298	U306	U314	U322	U330						
C	U291	U299	U307	U315	U323	U331						
D	U292	U300	U308	U316	U324	U332						
E	U293	U301	U309	U317	U325	U333						
F	U294	U302	U310	U318	U326	U334						
G	U295	U303	U311	U319	U327	U335						
H	U296	U304	U312	U320	U328	U336						

48-H Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U337	U345	U353	U361	U369	U377						
B	U338	U346	U354	U362	U370	U378						
C	U339	U347	U355	U363	U371	U379						
D	U340	U348	U356	U364	U372	U380						
E	U341	U349	U357	U365	U373	U381						
F	U342	U350	U358	U366	U374	U382						
G	U343	U351	U359	U367	U375	U383						
H	U344	U352	U360	U368	U376	U384						

For 64 Sample Kit

64-A Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U001	U009	U017	U025	U033	U041	U049	U057				
B	U002	U010	U018	U026	U034	U042	U050	U058				
C	U003	U011	U019	U027	U035	U043	U051	U059				
D	U004	U012	U020	U028	U036	U044	U052	U060				
E	U005	U013	U021	U029	U037	U045	U053	U061				
F	U006	U014	U022	U030	U038	U046	U054	U062				
G	U007	U015	U023	U031	U039	U047	U055	U063				
H	U008	U016	U024	U032	U040	U048	U056	U064				

64-B Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U065	U073	U081	U089	U097	U105	U113	U121				
B	U066	U074	U082	U090	U098	U106	U114	U122				
C	U067	U075	U083	U091	U099	U107	U115	U123				
D	U068	U076	U084	U092	U100	U108	U116	U124				
E	U069	U077	U085	U093	U101	U109	U117	U125				
F	U070	U078	U086	U094	U102	U110	U118	U126				
G	U071	U079	U087	U095	U103	U111	U119	U127				
H	U072	U080	U088	U096	U104	U112	U120	U128				

64-C Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U129	U137	U145	U153	U161	U169	U177	U185				
B	U130	U138	U146	U154	U162	U170	U178	U186				
C	U131	U139	U147	U155	U163	U171	U179	U187				
D	U132	U140	U148	U156	U164	U172	U180	U188				
E	U133	U141	U149	U157	U165	U173	U181	U189				
F	U134	U142	U150	U158	U166	U174	U182	U190				
G	U135	U143	U151	U159	U167	U175	U183	U191				
H	U136	U144	U152	U160	U168	U176	U184	U192				

64-D Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U193	U201	U209	U217	U225	U233	U241	U249				
B	U194	U202	U210	U218	U226	U234	U242	U250				
C	U195	U203	U211	U219	U227	U235	U243	U251				
D	U196	U204	U212	U220	U228	U236	U244	U252				
E	U197	U205	U213	U221	U229	U237	U245	U253				
F	U198	U206	U214	U222	U230	U238	U246	U254				
G	U199	U207	U215	U223	U231	U239	U247	U255				
H	U200	U208	U216	U224	U232	U240	U248	U256				

64-E Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U257	U265	U273	U281	U289	U297	U305	U313				
B	U258	U266	U274	U282	U290	U298	U306	U314				
C	U259	U267	U275	U283	U291	U299	U307	U315				
D	U260	U268	U276	U284	U292	U300	U308	U316				
E	U261	U269	U277	U285	U293	U301	U309	U317				
F	U262	U270	U278	U286	U294	U302	U310	U318				
G	U263	U271	U279	U287	U295	U303	U311	U319				
H	U264	U272	U280	U288	U296	U304	U312	U320				

64-F Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U321	U329	U337	U345	U353	U361	U369	U377				
B	U322	U330	U338	U346	U354	U362	U370	U378				
C	U323	U331	U339	U347	U355	U363	U371	U379				
D	U324	U332	U340	U348	U356	U364	U372	U380				
E	U325	U333	U341	U349	U357	U365	U373	U381				
F	U326	U334	U342	U350	U358	U366	U374	U382				
G	U327	U335	U343	U351	U359	U367	U375	U383				
H	U328	U336	U344	U352	U360	U368	U376	U384				

For 96 Sample Kit

96-A Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U001	U009	U017	U025	U033	U041	U049	U057	U065	U073	U081	U089
B	U002	U010	U018	U026	U034	U042	U050	U058	U066	U074	U082	U090
C	U003	U011	U019	U027	U035	U043	U051	U059	U067	U075	U083	U091
D	U004	U012	U020	U028	U036	U044	U052	U060	U068	U076	U084	U092
E	U005	U013	U021	U029	U037	U045	U053	U061	U069	U077	U085	U093
F	U006	U014	U022	U030	U038	U046	U054	U062	U070	U078	U086	U094
G	U007	U015	U023	U031	U039	U047	U055	U063	U071	U079	U087	U095
H	U008	U016	U024	U032	U040	U048	U056	U064	U072	U080	U088	U096

96-B Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U097	U105	U113	U121	U129	U137	U145	U153	U161	U169	U177	U185
B	U098	U106	U114	U122	U130	U138	U146	U154	U162	U170	U178	U186
C	U099	U107	U115	U123	U131	U139	U147	U155	U163	U171	U179	U187
D	U100	U108	U116	U124	U132	U140	U148	U156	U164	U172	U180	U188
E	U101	U109	U117	U125	U133	U141	U149	U157	U165	U173	U181	U189
F	U102	U110	U118	U126	U134	U142	U150	U158	U166	U174	U182	U190
G	U103	U111	U119	U127	U135	U143	U151	U159	U167	U175	U183	U191
H	U104	U112	U120	U128	U136	U144	U152	U160	U168	U176	U184	U192

96-C Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U193	U201	U209	U217	U225	U233	U241	U249	U257	U265	U273	U281
B	U194	U202	U210	U218	U226	U234	U242	U250	U258	U266	U274	U282
C	U195	U203	U211	U219	U227	U235	U243	U251	U259	U267	U275	U283
D	U196	U204	U212	U220	U228	U236	U244	U252	U260	U268	U276	U284
E	U197	U205	U213	U221	U229	U237	U245	U253	U261	U269	U277	U285
F	U198	U206	U214	U222	U230	U238	U246	U254	U262	U270	U278	U286
G	U199	U207	U215	U223	U231	U239	U247	U255	U263	U271	U279	U287
H	U200	U208	U216	U224	U232	U240	U248	U256	U264	U272	U280	U288

96-D Set

	1	2	3	4	5	6	7	8	9	10	11	12
A	U289	U297	U305	U313	U321	U329	U337	U345	U353	U361	U369	U377
B	U290	U298	U306	U314	U322	U330	U338	U346	U354	U362	U370	U378
C	U291	U299	U307	U315	U323	U331	U339	U347	U355	U363	U371	U379
D	U292	U300	U308	U316	U324	U332	U340	U348	U356	U364	U372	U380
E	U293	U301	U309	U317	U325	U333	U341	U349	U357	U365	U373	U381
F	U294	U302	U310	U318	U326	U334	U342	U350	U358	U366	U374	U382
G	U295	U303	U311	U319	U327	U335	U343	U351	U359	U367	U375	U383
H	U296	U304	U312	U320	U328	U336	U344	U352	U360	U368	U376	U384